GIT & GITHUB

1. Install Git

1.1 Download Git

For Windows / macOS / Linux:

1. Open browser → search **"Download Git"**
2. Open official site → **https://git-scm.com**
3. Click **Download** for your OS:
   - **Windows** → Git for Windows (.exe)
   - **macOS** → .dmg installer (or install via Homebrew)
   - **Linux** → use package manager:
     Ubuntu / Debian: `sudo apt install git`
     Fedora: `sudo dnf install git`
     Arch: `sudo pacman -S git`

1.2 Verify Git Installation

Open CMD / Terminal / PowerShell:

```
git --version
```

If version appears → Git is installed correctly.

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2 Clone Repository

```
git clone https://github.com/your-username/repo-name.git
cd repo-name
```

Repository is now on your machine.

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3. Create GitHub Account

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1. Open **https://github.com**
2. Click **Sign Up**
3. Enter:
    - Email
    - Password
    - Username
4. Verify email using the OTP sent to inbox.
5. Complete basic profile setup.

Your GitHub account is ready.

3.1 Configure Git Identity (Only Once)

```
git config --list

git config --global user.name "Your Name"

git config --global user.email "youremail@example.com"
```

---

4. Enable Two-Factor Authentication (2FA) with Authenticator App

GitHub strongly recommends 2FA for account security.

4.1 Open GitHub Security Settings

1. Log into GitHub
2. Click profile → **Settings**
3. Go to **Password and authentication**

4.2 Start 2FA Setup

1. **Enable two-factor authentication**
2. Select **Authentication app** (recommended method)
3. GitHub will display:
    - QR code
    - Secret key

4.3 Configure Chrome Authenticator Extension

1. Open Chrome Web Store
2. Search → **Authenticator**
3. Click **Add to Chrome** → **Add extension**
4. Open the extension
5. Scan GitHub's QR code OR enter secret key manually

The extension will generate 6-digit OTPs.

4.4 Complete 2FA Setup

1. Enter the 6-digit code from the Authenticator extension
2. Save GitHub's **recovery codes** safely

2FA is now enabled and secure.

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5. Create Private GitHub Organisation

Naming Format:

```
CourseName-FirstName-RollNumber
```

Example:

```
AC-Yogesh-88333
```

Steps:

```
1. Profile → Your Organizations

2. New Organization

3. Select Free Plan

4. Enter organisation name

5. Complete setup
```

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5.1 Create Two Repositories for each module

Inside organisation create:

1. Assignments (Private)
2. Self-Study (Private)

Initialize both with README.

5.2 Module-wise Folder Structure

Assignments:

```
C++ Assignments/
│
├── Assignments-01/
│
├── Assignments-02/
│
├── Assignments-03/
│
└── README.md
```

Self-Study

```
C++ Self-Study/
│
├── Assignments-01/
│
├── Assignments-02/
│
├── Assignments-03/
│
└── README.md
```

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6. Inviting A Collaborator

Step 1: Open Organisation

1. Log in to GitHub.

2. Click your Profile Icon (Top Right).
3. Click Your Organizations.
4. Select your organisation (e.g., AC_Yogesh_88333).

Step 2: Open Organisation Settings

1. Inside organisation dashboard.
2. Click Settings.
3. Click People from left sidebar.

Step 3: Invite Member

1. Click Invite member button.
2. Enter:
    - GitHub username OR
    - Registered email address
3. Click Invite.

Step 4: Assign Role

During invitation, choose role:

- Owner → Full control (use carefully)
- Member → Standard access (recommended)

Click Send Invitation.

Step 5: Collaborator Accepts Invitation

1. Collaborator receives email notification.
2. OR sees notification inside GitHub.
3. Click Accept Invitation.

Once accepted, member becomes part of organisation.

7. Grant Access to Specific Repository

After adding member to organisation, you must assign repository access.

Step 1: Open Repository

1. Go to organisation.
2. Click repository (e.g., Assignments or Self-Study).
3. Click Settings.

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Step 2: Manage Access

1. Click Collaborators and Teams (or Manage Access).
2. Click Add people.
3. Select organisation member.
4. Choose permission level:

Permissions Available:

- Read → Can view repository
- Triage → Manage issues
- Write → Push code
- Maintain → Manage repository settings
- Admin → Full control

Recommended for students: → Write Access

Click Add.

8. Verify Access

After adding:

- Collaborator can clone repository.
- Collaborator can create branches.
- Collaborator can push changes (if Write permission given).
- Collaborator can create Pull Requests.

9. Remove Member (If Required)

Remove from Repository

1. Open Repository → Settings.
2. Manage Access.
3. Remove collaborator.

Remove from Organisation

1. Organisation → Settings.
2. People.
3. Click 3 dots next to member name.
4. Select Remove from organisation.

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10. Steps to Create Personal Access Token (Classic)

Step 1: Login to GitHub

Open: https://github.com

Login with your credentials.

Step 2: Open Developer Settings

1. Click Profile icon (top right)
2. Click Settings
3. Scroll down left sidebar
4. Click Developer settings

Step 3: Open Personal Access Tokens

1. Click Personal access tokens
2. Click Tokens (classic)

Step 4: Generate New Token

1. Click Generate new token
2. Select Generate new token (classic)

Step 5: Configure Token Settings

Fill the following details:

Token Name (Note)

Example: AC_Assignments_Token

Expiration

Choose appropriate duration:

- 7 days
- 30 days
- 90 days

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- No expiration (Not recommended for security)

Recommended: Set limited expiration.

Step 6: Select Scopes (Permissions)

Choose required permissions carefully.

Commonly Required Scopes:

```
✔ repo - Full control of private repositories
✔ workflow - Update GitHub Actions workflows
✔ read:org - Read organisation data
```

For pushing code to private repositories: Select: repo

Avoid selecting unnecessary permissions.

Step 7: Generate Token

1. Click Generate token
2. GitHub will display the token ONLY ONCE

IMPORTANT: Copy the token immediately and store it securely.

Example token format: ghp_xxxxxxxxxxxxxxxxxxxxxxxxxxxxxx

If you lose it → You must regenerate a new token.

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11. Make Changes: Add → Commit → Push

8.1 Check Status

git status

8.2 Stage Files Add a specific file:

git add file.txt Add all changes: git add .

8.3 Commit Changes

git commit -m "Add login feature"

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8.4 Push Branch to GitHub

git push

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12. How to Use Personal Access Token

When pushing code using HTTPS:

Example:

```
git clone https://github.com/ORG_NAME/Assignments.git
```

When prompted:

Username: Your GitHub Username

Password: Paste Personal Access Token (NOT your GitHub password)

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